

StockSense AI

Retail sales forecasting and stock planning

VUT AI Solutions / Business Analysis 3.2

A fictional retail case

A manager needs a consistent estimate before deciding how much to order. StockSense shows seven daily sales estimates and explains a suggested quantity using entered stock and delivery assumptions.

One store. Five items. Seven forecast days.

Kaggle anonymous sales records, 2013-2017. Strict checks select 9,130 rows. The local Python dashboard combines sales history, forecasts, a stock calculator and a five-intent text assistant.

Measured historical comparison

2017 final test

Method	Daily MAE	Daily RMSE	Weekly-total MAE
Weekly benchmark	6.273	8.218	18.427
Random forest	4.812	6.387	17.973
Small LSTM	5.411	7.261	17.343

Errors are in units. Random forest has about 23.3% lower daily MAE than the benchmark. LSTM has lower overall weekly-total error. Selection used 2016 daily MAE and stayed frozen before final testing.

A transparent stock decision

Coverage = lead time + review interval, up to seven days.
Target = forecast over coverage + buffer.
Order = round upwards max(0, target - stock - outstanding units).
The screen warns when stock may run out before delivery.

Verified on the personal PC

45 automated tests passed, including all five items and the stock calculator. The assistant recognised 24/25 supported test requests and rejected 8/8 unrelated requests.

Historical demonstration only. January 2018 estimates have no observed outcomes. No real savings or stockout reduction has been established. Stock and supplier inputs are demonstration assumptions.